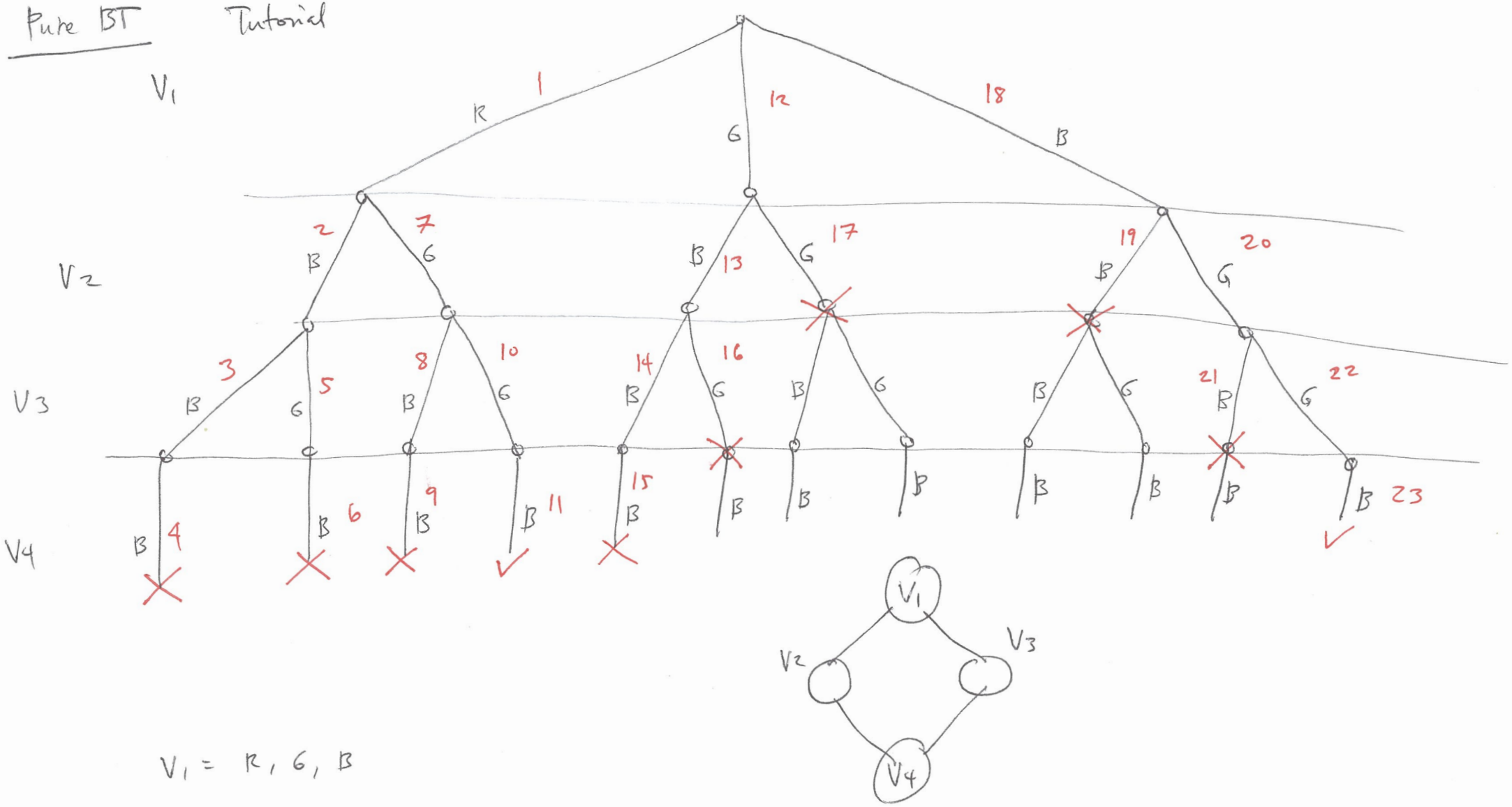
BI Tutorial #9

1) 1R
2) 2B
3) 3B
4) 4B
5) 3G
6) 4B
7) 2G
8) 3B
9) 4B
10) 3G
11) 4B
12) 1G
13) 2B
14) 3B
15) 4B
16) 3G
17) 2G
18) 4B
19) 2B
20) 2G
21) 3B
22) 3G
23) 4B

Pure BT Tutorial



V_1 = R, G, B

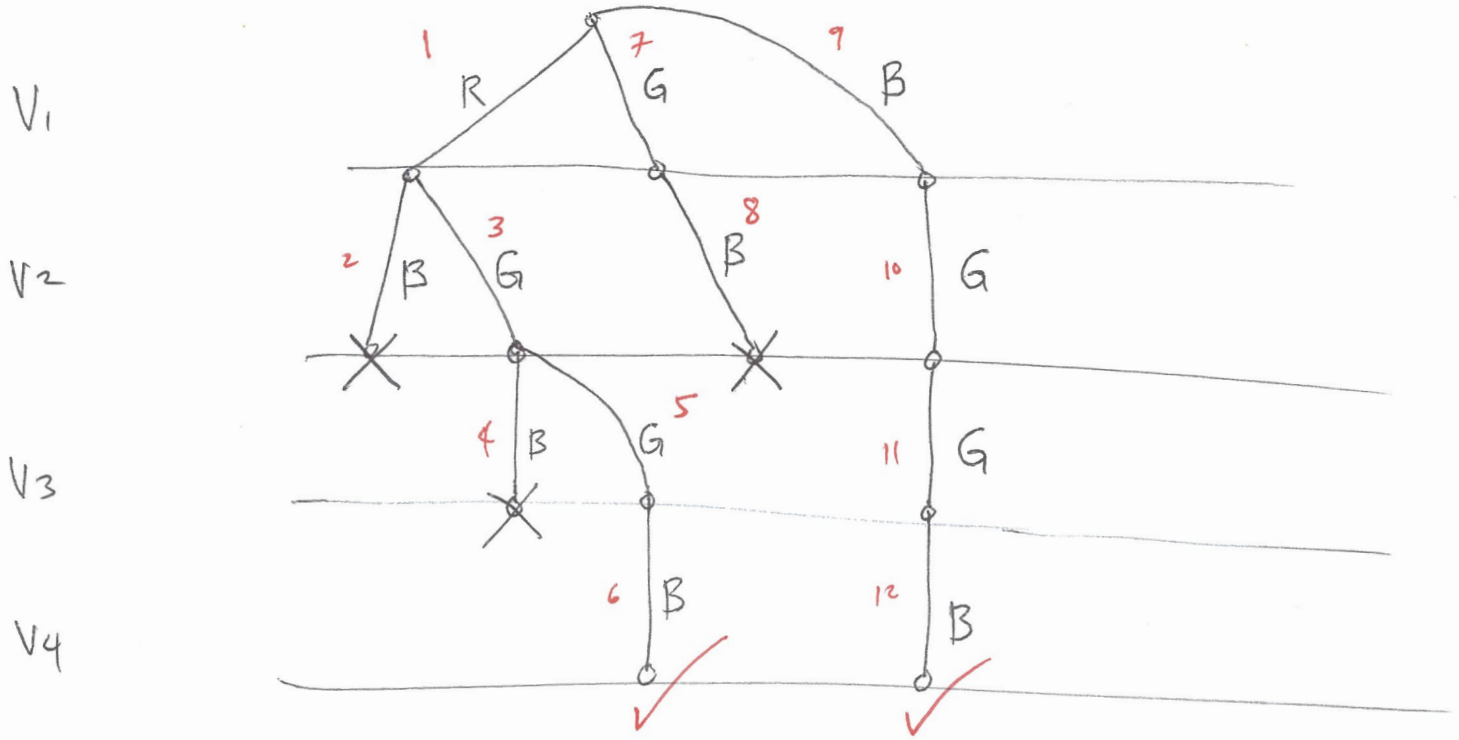
V_2 = B, G

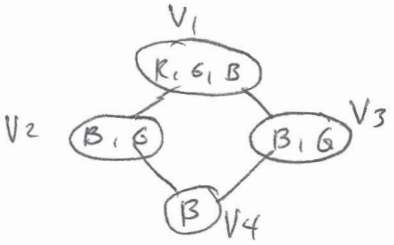
V_3 = B, G

V_4 = B

Tutorial

BT-FC





BT-FC

1. $V_1 = \textcircled{R}\ G\ B$
   $V_2 = B\ G$
   $V_3 = B\ G$
   $V_4 = B$

2. $V_1 = \textcircled{R}\ G\ B$
   $V_2 = \textcircled{B}\ G$
   $V_3 = B\ G$
   $V_4 = \cancel{B}$ empty
   no soln.

3. $V_1 = \textcircled{R}\ G\ B$
   $V_2 = \textcircled{G}$
   $V_3 = B, G$
   $V_4 = B$

4. $V_1 = \textcircled{R}\ G\ B$
   $V_2 = \textcircled{G}$
   $V_3 = \textcircled{B}\ G$
   $V_4 = \cancel{B}$ empty
   no soln.

5. $V_1 = \textcircled{R}\ G\ B$
   $V_2 = \textcircled{G}$
   $V_3 = \textcircled{G}$
   $V_4 = B$

6. $V_1 = \textcircled{R}\ G\ B$
   $V_2 = \textcircled{G}$
   $V_3 = \textcircled{G}$
   $V_4 = \textcircled{B}$
   ✓ 1 soln. found

7. $V_1 = \textcircled{G}\ B$
   $V_2 = B\ \cancel{G}$
   $V_3 = B\ \cancel{G}$
   $V_4 = B$

8. $V_1 = \textcircled{G}\ B$
   $V_2 = \textcircled{B}$
   $V_3 = B$
   $V_4 = \cancel{B}$ empty
   no soln.

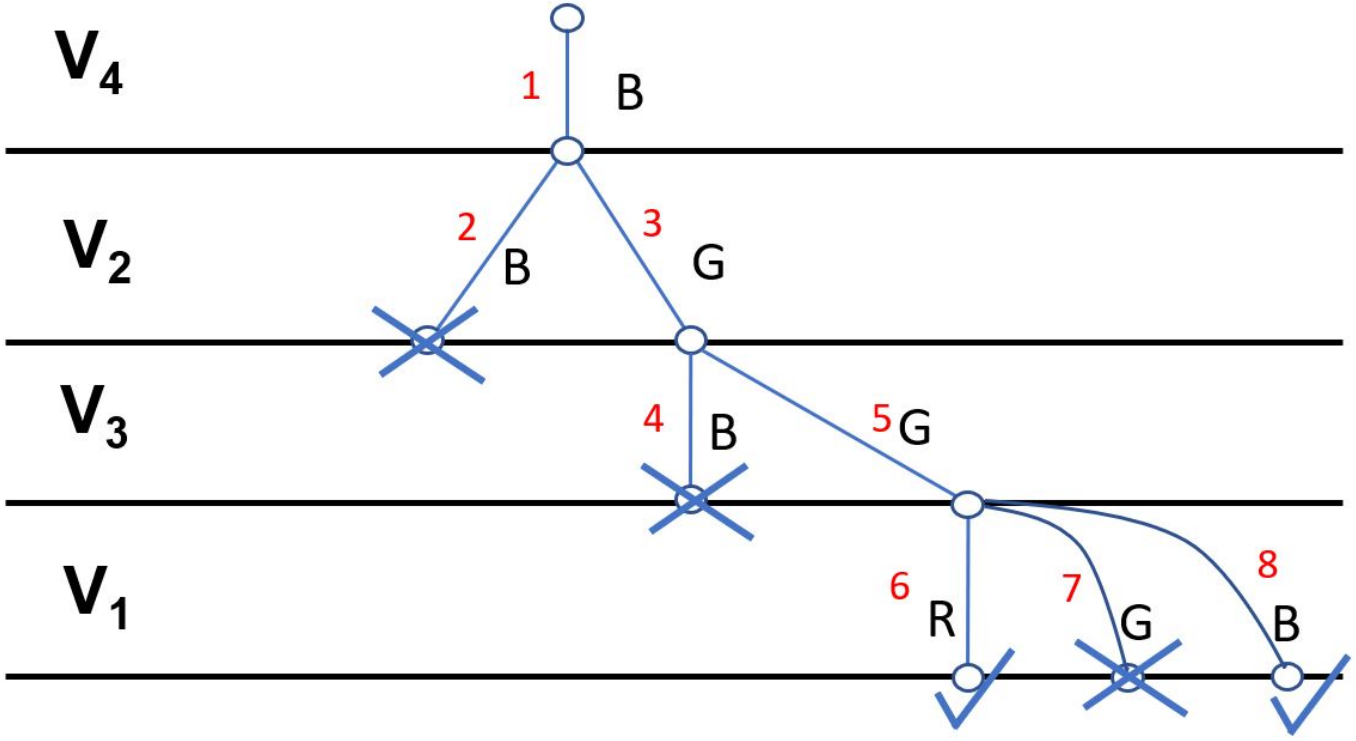
9. $V_1 = \textcircled{B}$
   $V_2 = \cancel{B}, G$
   $V_3 = \cancel{B}, G$
   $V_4 = B$

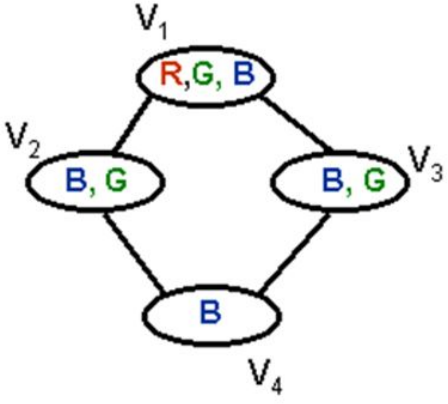
10. $V_1 = \textcircled{B}$
    $V_2 = \textcircled{G}$
    $V_3 = G$
    $V_4 = B$

11. $V_1 = \textcircled{B}$
    $V_2 = \textcircled{G}$
    $V_3 = \textcircled{G}$
    $V_4 = B$

12. $V_1 = \textcircled{B}$
    $V_2 = \textcircled{G}$
    $V_3 = \textcircled{G}$
    $V_4 = \textcircled{B}$
    ✓ 2nd solution found

Tutorial: Backtracking with Dynamic Variable Ordering





BT – Dynamic Variable Ordering

0. MCV ordering V_4, V_2, V_3, V_1

1. $V_4 = B$

2. $V_2 = B$ $V_2 = G$

$V_1 = R, G, B$ $V_1 = R, G, B$

rules out 1 rules out 1 value

$V_2 = B$, conflicts with $V_4 = B$

3. $V_2 = G$

4. V3 = B V3 = G

V1 = R, G, B V1 = R, G, B

Rules out 1 Rules out 1

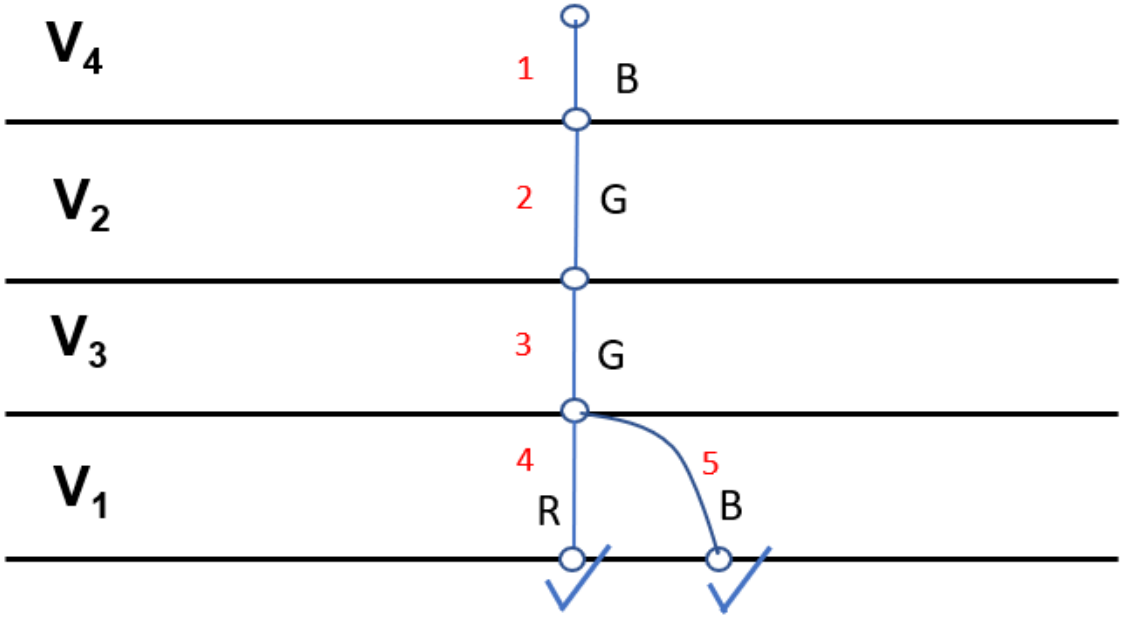
5. V3 = G

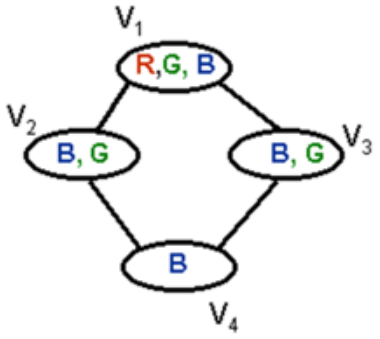
6. V1 = R, G, B

7. V1 = G, B

8. V1 = B

Tutorial: Backtracking with Forward Checking and Dynamic Variable Ordering





BT-FC Dynamic Variable Ordering

0. MCV Ordering : V_4, V_2, V_3, V_1

1. $V_4 = \{B\}$

   $V_2 = \cancel{B}, G$

   $V_3 = \cancel{B}, G$

   $V_1 = R, G, \cancel{B}$

2. $V_4 = \{B\}$

   $V_2 = \{G\}$

   $V_3 = G$

   $V_1 = R, \cancel{G}, B$

3. $V_4 = \{B\}$

   $V_2 = \{G\}$

   $V_3 = \{G\}$

   $V_1 = R, B$

4. $V_4 = \{B\}$

   $V_2 = \{G\}$

   $V_3 = \{G\}$

   $V_1 = \{R\}, B$

1st solution found!

5. $V_4 = \{B\}$

   $V_2 = \{G\}$

   $V_3 = \{G\}$

   $V_1 = \{B\}$

2nd solution found!